Automated Pneumatic Pick-and-Place System with HMI Monitoring and Control

Technological University of the Philippines-Visayas

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Instructor in HMI Systems

(MxE 412A)

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Pneumatic Pick-and-Place System

HMI Systems (MxE 412A)

Bachelor of Science in Mechatronics Engineering

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BS MxE 4A

2. Project Overview

2.1 Purpose of the Machine/Process

The project involves the development of a PLC-controlled automated pneumatic pick-and-place station designed to demonstrate an automated material-handling process. The system performs a sequential pick-and-place operation using pneumatic actuators and a pneumatic gripper. The machine is capable of moving the gripper horizontally and vertically to pick up and transfer a workpiece between designated positions.

The system is operated and monitored through a Human-Machine Interface (HMI), allowing the operator to observe the machine status, actuator positions, and system conditions.

2.2 Training/Industrial Application

The system is intended primarily as an industrial automation training application. It demonstrates concepts commonly used in automated material-handling systems, including PLC programming, pneumatic control, sensor feedback, sequence control, HMI operation, alarm handling, and safety functions.

2.3 PLC/Controller Used

The system utilizes a Siemens S7 200CN SMART1 as the main controller. The PLC processes input signals from the sensors and operator controls and generates output signals for the pneumatic actuators and other controlled devices.

2.4 Sensors and Actuators Used

The system uses position/proximity sensors to detect the position of the pneumatic mechanisms. Pneumatic cylinders are used to provide horizontal and vertical movement, while a pneumatic gripper is used to grasp and release the workpiece.

The primary devices include:

- Reed Switch
- Horizontal pneumatic actuator
- Vertical pneumatic actuator
- Pneumatic gripper
- Solenoid valves
- Operator pushbuttons/control inputs

2.5 Material or Product Handled

The system is designed to handle a small 3d printed workpiece during the pick-and-place sequence, but the simulation was done without using a piece and just simulating the sequence of the machine station.

2.6 Operating Mode

The system operates in Manual Push Button mode. In automatic operation, the PLC executes the programmed pick-and-place sequence based on sensor feedback and predefined conditions.

2.7 HMI/SCADA Software Used

A Haiwell Cloud SCADA Designer was used to provide operator control and real-time monitoring of the station. The interface includes an overview screen, control and monitoring screen, 2D machine visualization, and alarm/system history.

3. System Architecture and Block Diagram

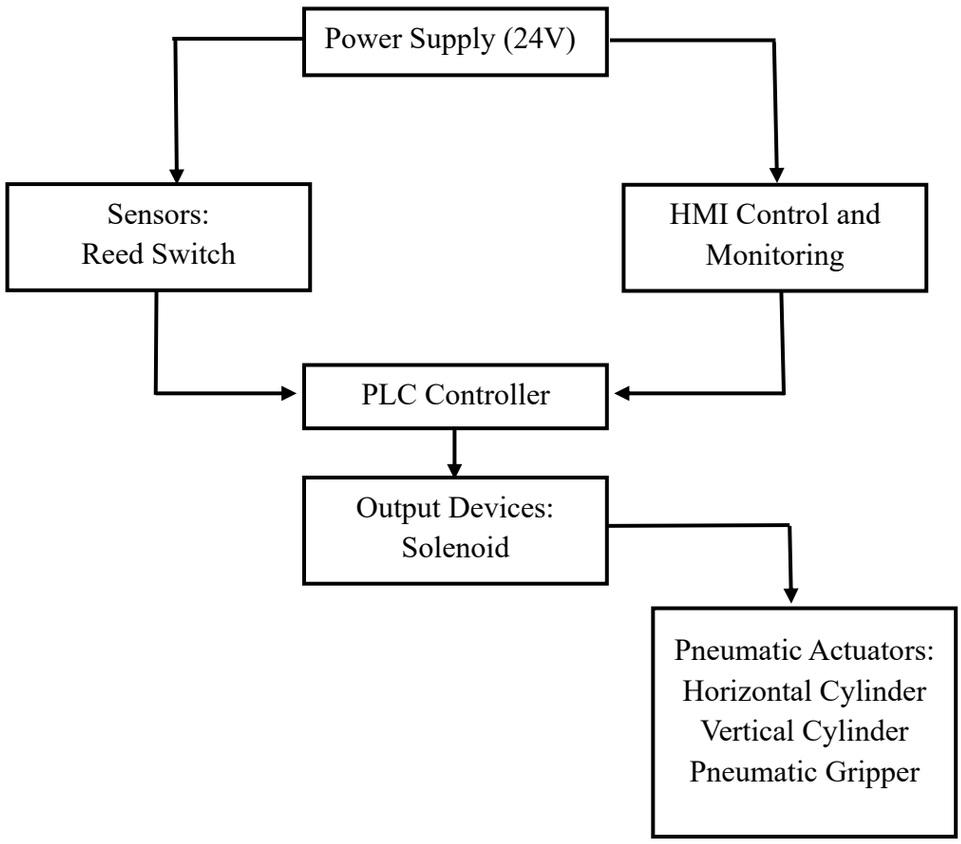
3.1 System Architecture

The system consists of a power supply, PLC controller, input sensors, pneumatic output devices, and HMI. The PLC serves as the central control unit, receiving signals from the sensors and operator controls and generating commands for the pneumatic actuators.

The HMI communicates with the PLC to provide operator control and real-time visualization of the machine status.

The pneumatic section consists of the compressed-air supply, air preparation components, solenoid valves, pneumatic cylinders, and gripper.

3.2 System Architecture



4. User Requirements Specification (URS)

4.1 Start/Stop Requirement

The system shall provide operator controls for starting and stopping the automated pick-and-place sequence. The START command shall initiate the sequence when the required operating conditions are satisfied, while the STOP command shall stop the automated operation and return the system to the appropriate safe state.

4.2 Monitoring Requirement

The HMI shall provide real-time monitoring of the machine operating condition, including system state, actuator position, gripper condition, and sensor status.

4.3 Alarm Requirement

The system shall provide alarm indications for Emergency Stops triggered. Alarm information shall be presented through the HMI.

4.4 Safety/Emergency Stop Requirement

An Emergency Stop function shall be provided to immediately interrupt machine operation in the event of an emergency or unsafe condition.

4.5 Control Requirement

The operator shall be able to control the machine through the HMI. The control interface shall include the necessary operating commands, such as START, STOP, and PAUSE.

4.6 Alarm History Requirement

The HMI shall provide an alarm/system history interface for viewing previously detected alarms and system events, including their corresponding time or status when supported by the system.

4.7 Communication Requirement

The HMI and PLC shall communicate through **Ethernet (TCP/IP)** to exchange control commands and monitoring data.

5. Functional Design Specification (FDS)

5.1 Digital Input Functions

Digital inputs are used to receive signals from operator controls and machine sensors. These inputs include the START, STOP, PAUSE, as well as position/proximity sensors used to determine the location of the pneumatic mechanisms.

5.2 Digital Output Functions

Digital outputs from the PLC control the solenoid valves and other actuators responsible for the movement of the pick-and-place mechanism.

The outputs control:

- Horizontal movement

- Vertical movement
- Gripper operation
- Status indicators, where applicable

5.3 Automatic Mode Function

In automatic mode, the PLC executes the predefined pick-and-place sequence.

A simplified sequence is:

System Ready?

↓

Start (Manual Press of External Buttons)

↓

Move to Pick Position

↓

Lower / Extend

↓

Grip Workpiece

↓

Raise / Retract

↓

Move to Place Position

↓

Lower / Extend

↓

Release Workpiece

↓

Return to Home Position

↓

STANDBY / Cycle Complete

5.4 HMI/SCADA Control Functions

The HMI shall provide the operator with control and monitoring functions for the automated station.

The HMI shall contain the following primary screens:

A. Project Overview

Displays the station overview, machine representation, and overall operating status.

B. Controls & Monitoring

Provides operator controls including:

- START
- STOP
- PAUSE
- E-STOP

and displays system states such as:

- STANDBY
- RUNNING
- PAUSED
- SYSTEM HALT

C. 2D Monitoring

Provides a graphical representation of the pick-and-place mechanism and displays real-time position and actuator feedback.

The visualization includes:

- Left position
- Right position
- Retracted position
- Extended position
- Gripper status
- Sensor feedback

D. Alarm System History

Displays active and/or historical alarms and system events generated during operation.

5.6 Alarm Functions

The system shall detect and indicate abnormal operating conditions through the HMI. Alarm priority shall follow the specified HMI color-coding standard.

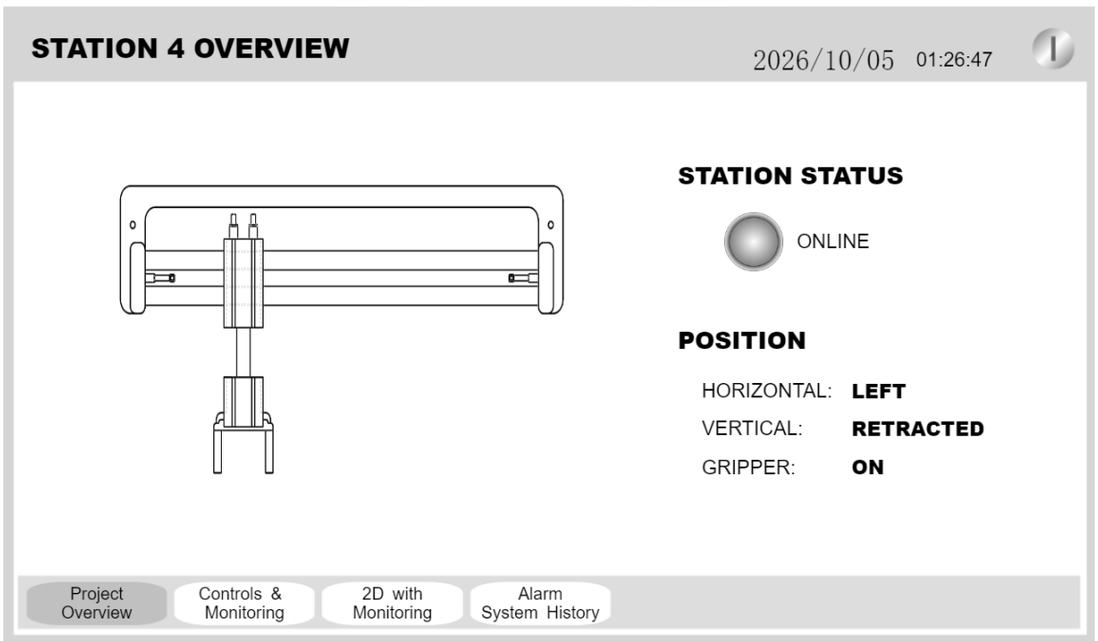
Priority	**Color**	**Example**
Priority 1	**Red**	Emergency Stop Alarm

5.7 HMI Design

5.7.1 Project Overview Screen

Figure 5.1. Project Overview HMI Screen

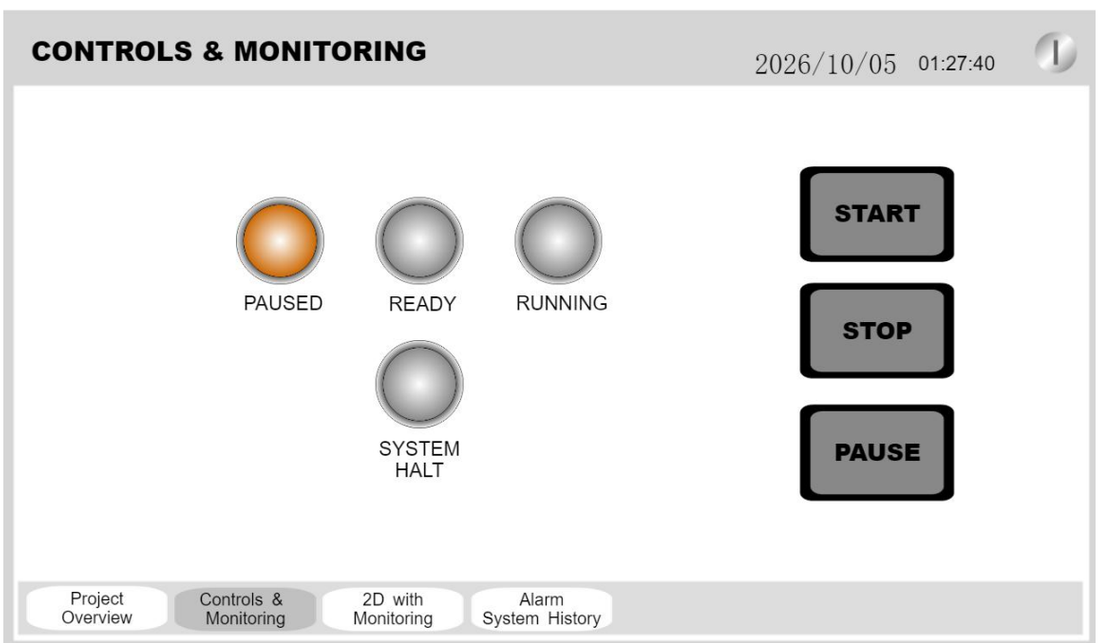
The Project Overview screen provides a summary of the Station 4 pick-and-place system. The screen displays the machine visualization, station operating status, date and time, and navigation controls to the other HMI screens.



5.7.2 Controls and Monitoring Screen

Figure 5.2. Controls and Monitoring HMI Screen

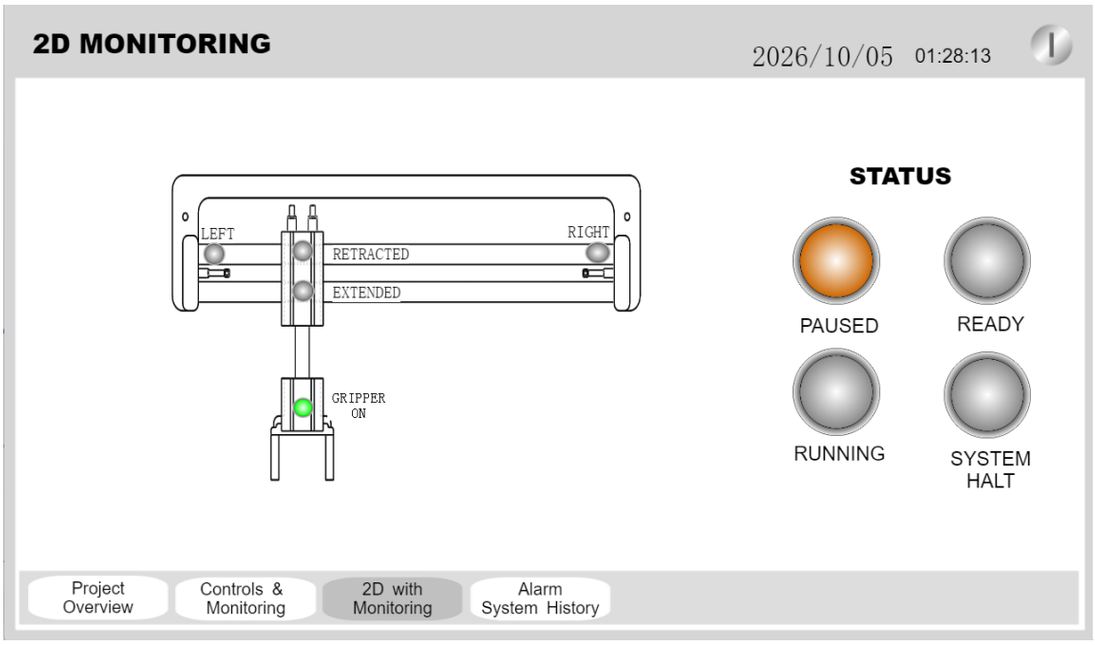
The Controls and Monitoring screen provides the operator with direct control of the station through START, STOP, PAUSE, and Emergency Stop functions. The screen also displays the current operating state of the system.



5.7.3 2D Monitoring Screen

Figure 5.3. 2D Monitoring HMI Screen

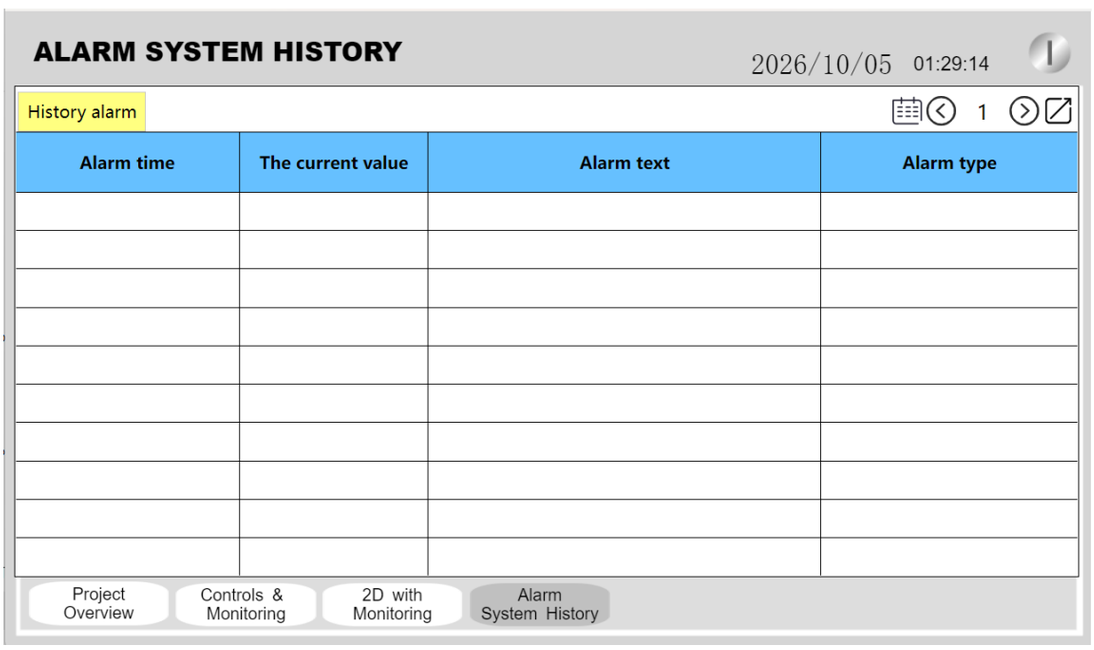
The 2D Monitoring screen provides a graphical representation of the pneumatic pick-and-place mechanism. Position feedback from the sensors is displayed on the machine visualization, allowing the operator to identify the current horizontal and vertical position of the mechanism and the state of the gripper.



5.7.4 Alarm System History Screen

Figure 5.4. Alarm System History HMI Screen

The Alarm System History screen displays system alarms about emergency stops triggered during operation. Alarm indications are categorized according to their assigned priority and the specified HMI color-coding standard.



PLC I/O and Tag Database

1. Digital Inputs

These are the physical inputs going into the PLC.

Address	Tag Name	Device / Signal	Function
I0.0	RS_1_X	X-Axis Position Sensor 1	Detects X-axis position 1
I0.1	RS_2_X	X-Axis Position Sensor 2	Detects X-axis position 2
I0.2	RS_4_Y	Y-Axis Position Sensor 1	Detects Y-axis position
I0.3	RS_5_GRIPPER	Gripper Reed Switch	Detects gripper position/state
I0.5	RS_3_Y	Y-Axis Position Sensor 2	Detects Y-axis position
I1.0	START	Start Pushbutton	Initiates system operation
I1.1	RESET	Reset Pushbutton	Resets system/fault state
I1.2	SPECIAL	Special Pushbutton	Special/manual control function
I1.5	STOP_	Stop Pushbutton	Stops system operation
I1.6	SCRAM	Emergency Stop	Immediately halts system operation

2. Digital Outputs

Address	Tag Name	Device / Actuator	Function
Q0.2	X_AXIS_1	X-Axis Solenoid Valve 1	Controls X-axis movement in one direction
Q0.3	X_AXIS_2	X-Axis Solenoid Valve 2	Controls X-axis movement in the opposite direction
Q0.4	GRIPPER	Gripper Solenoid Valve	Controls pneumatic gripper
Q0.5	Y_AXIS	Y-Axis Solenoid Valve	Controls Y-axis pneumatic movement

3. Internal PLC Tags

Address	Tag Name	Function
M0.0	ON_STATE	Indicates that the system is enabled/ON
M0.1	PROCESS	Indicates that the process has started
M0.2	X_AXIS_STATE	Stores/indicates X-axis operating state
M0.3	Y_AXIS_STATE	Stores/indicates Y-axis operating state
M0.4	GRIPPER_STATE	Stores/indicates gripper state
M0.5	PROGRAM_STARTED	Indicates that the program has been started
M0.6	READY_STATE	Indicates that the system is ready
M0.7	PAUSE_STATE	Indicates that the system is paused
M4.0	RUNNING_STATE	Indicates that the system is currently running

4. Automatic Sequence Tags

Address	Tag Name	Function
M1.0	STEP_1_DONE	Indicates completion of Automatic Sequence Step 1
M1.1	STEP_2_DONE	Indicates completion of Automatic Sequence Step 2
M1.2	STEP_3_DONE	Indicates completion of Automatic Sequence Step 3
M1.3	STEP_4_DONE	Indicates completion of Automatic Sequence Step 4

M1.4	**STEP_5_DONE**	**Indicates completion of Automatic Sequence Step 5**
M1.5	**STEP_6_DONE**	**Indicates completion of Automatic Sequence Step 6**
M1.6	**STEP_7_DONE**	**Indicates completion of Automatic Sequence Step 7**
M1.7	**STEP_8_DONE**	**Indicates completion of Automatic Sequence Step 8**

5. HMI Command Tags

Address	Tag Name	HMI Function
M2.0	**HMI_START**	**HMI Start command**
M2.1	**HMI_PAUSE**	**HMI Pause command**
M2.2	**HMI_STOP**	**HMI Stop command**
M2.3	**HMI_E_STOP**	**HMI Emergency Stop command**

Hardware Design

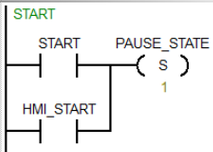
1. Complete Component List

No.	Component	Quantity	Function	Manufacturer / Model
1	PLC	1	Main controller; processes inputs and controls outputs	**Siemens S7 200CN SMART1**
2	HMI	1	Operator control and machine monitoring	**HAIWELL**
3	DC Power Supply	1	Provides DC power to control components	**N/A**
4	REED SWITCH	5	Detects actuator/carriage position	**N/A**
5	Solenoid Valve	**N/A**	Controls compressed air to pneumatic actuators	**N/A**
6	Horizontal Pneumatic Cylinder	1	Provides horizontal LEFT/RIGHT movement	**N/A**
7	Vertical Pneumatic Cylinder	1	Provides vertical EXTEND/RETRACT movement	**N/A**
8	Pneumatic Gripper	1	Grasps and releases the workpiece	**N/A**
9	Pushbutton	5	Provides operator commands	**N/A**
10	Emergency Stop	1	Provides emergency shutdown command	**N/A**
11	Terminal Block	**N/A**	Provides electrical connection/distribution	**N/A**
12	Relay	**N/A**	Interfaces/isolate control signals	**N/A**

PLC Software Documentation

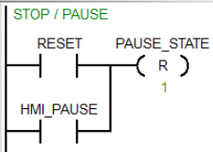
Program Comments

1 START



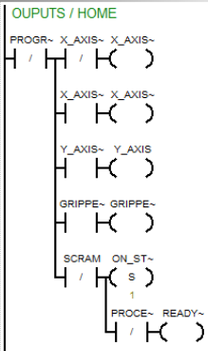
Symbol	Address	Comment
HMI_START	M2.0	HMI_START
PAUSE_STATE	M0.7	PAUSE_STATE
START	I1.0	START BUTTON

2 STOP / PAUSE



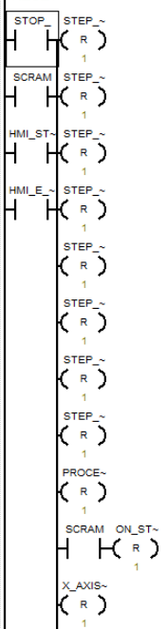
Symbol	Address	Comment
HMI_PAUSE	M2.1	HMI_PAUSE
PAUSE_STATE	M0.7	PAUSE_STATE
RESET	I1.1	RESET BUTTON

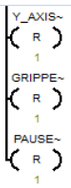
3 OUPUTS / HOME



Symbol	Address	Comment
GRIPPER	Q0.4	GRIPPER
GRIPPER_STATE	M0.4	GRIPPER_STATE
ON_STATE	M0.0	ON STATE
PROCESS	M0.1	PROCESS START
PROGRAM_STA...	M0.5	PROGRAM STARTED
READY_STATE	M0.6	READY_STATE
SCRAM	I1.6	EMERGENCY STOP
X_AXIS_1	Q0.2	X_AXIS_1
X_AXIS_2	Q0.3	X_AXIS_2
X_AXIS_STATE	M0.2	X_AXIS_STATE
Y_AXIS	Q0.5	Y_AXIS
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

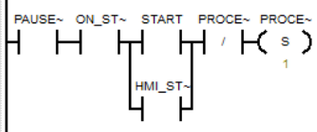
4 RESET





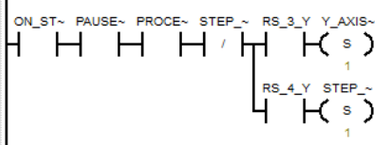
Symbol	Address	Comment
GRIPPER_STATE	M0.4	GRIPPER_STATE
HMI_E_STOP	M2.3	HMI_E_STOP
HMI_STOP	M2.2	HMI_STOP
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
PROCESS	M0.1	PROCESS START
SCRAM	I1.6	EMERGENCY STOP
STEP_1_DONE	M1.0	STEP_1_DONE
STEP_2_DONE	M1.1	STEP_2_DONE
STEP_3_DONE	M1.2	STEP_3_DONE
STEP_4_DONE	M1.3	STEP_4_DONE
STEP_5_DONE	M1.4	STEP_5_DONE
STEP_6_DONE	M1.5	STEP_6_DONE
STEP_7_DONE	M1.6	STEP_7_DONE
STEP_8_DONE	M1.7	STEP_8_DONE
STOP_	I1.5	STOP BUTTON
X_AXIS_STATE	M0.2	X_AXIS_STATE
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

5 PROCESS START



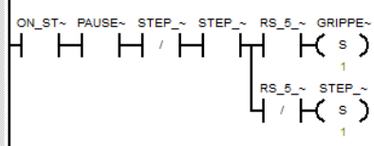
Symbol	Address	Comment
HMI_START	M2.0	HMI_START
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
PROCESS	M0.1	PROCESS START
START	I1.0	START BUTTON

6 STEP 1



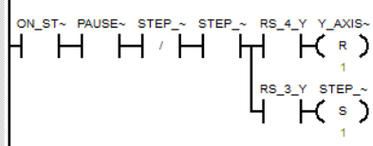
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
PROCESS	M0.1	PROCESS START
RS_3_Y	I0.5	REED SWITCH 3 Y
RS_4_Y	I0.2	REED SWITCH 4 Y
STEP_1_DONE	M1.0	STEP_1_DONE
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

7 STEP 2



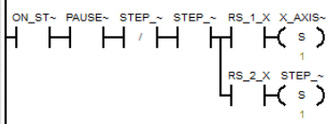
Symbol	Address	Comment
GRIPPER_STATE	M0.4	GRIPPER_STATE
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_5_GRIPPER	I0.3	REED SWITCH 5 GRIPPER
STEP_1_DONE	M1.0	STEP_1_DONE
STEP_2_DONE	M1.1	STEP_2_DONE

8 STEP 3



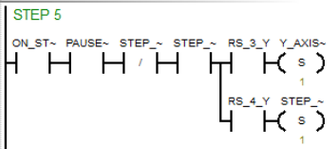
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_3_Y	I0.5	REED SWITCH 3 Y
RS_4_Y	I0.2	REED SWITCH 4 Y
STEP_2_DONE	M1.1	STEP_2_DONE
STEP_3_DONE	M1.2	STEP_3_DONE
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

9 STEP 4



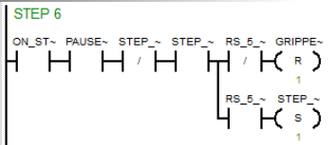
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_1_X	I0.0	REED SWITCH 1 X
RS_2_X	I0.1	REED SWITCH 2 X
STEP_3_DONE	M1.2	STEP_3_DONE
STEP_4_DONE	M1.3	STEP_4_DONE
X_AXIS_STATE	M0.2	X_AXIS_STATE

10 STEP 5



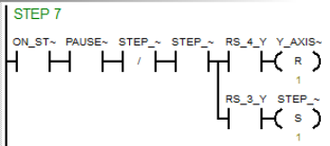
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_3_Y	I0.5	REED SWITCH 3 Y
RS_4_Y	I0.2	REED SWITCH 4 Y
STEP_4_DONE	M1.3	STEP_4_DONE
STEP_5_DONE	M1.4	STEP_5_DONE
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

11 STEP 6



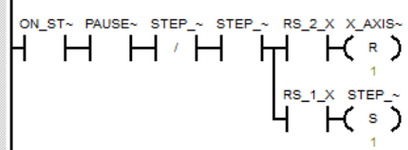
Symbol	Address	Comment
GRIPPER_STATE	M0.4	GRIPPER_STATE
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_5_GRIPPER	I0.3	REED SWITCH 5 GRIPPER
STEP_5_DONE	M1.4	STEP_5_DONE
STEP_6_DONE	M1.5	STEP_6_DONE

12 STEP 7



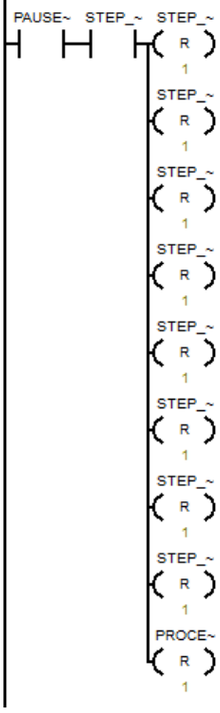
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_3_Y	I0.5	REED SWITCH 3 Y
RS_4_Y	I0.2	REED SWITCH 4 Y
STEP_6_DONE	M1.5	STEP_6_DONE
STEP_7_DONE	M1.6	STEP_7_DONE
Y_AXIS_STATE	M0.3	Y_AXIS_STATE

13 STEP 8



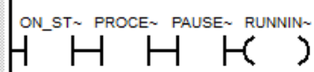
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
RS_1_X	I0.0	REED SWITCH 1 X
RS_2_X	I0.1	REED SWITCH 2 X
STEP_7_DONE	M1.6	STEP_7_DONE
STEP_8_DONE	M1.7	STEP_8_DONE
X_AXIS_STATE	M0.2	X_AXIS_STATE

14 RESET SEQUENCE



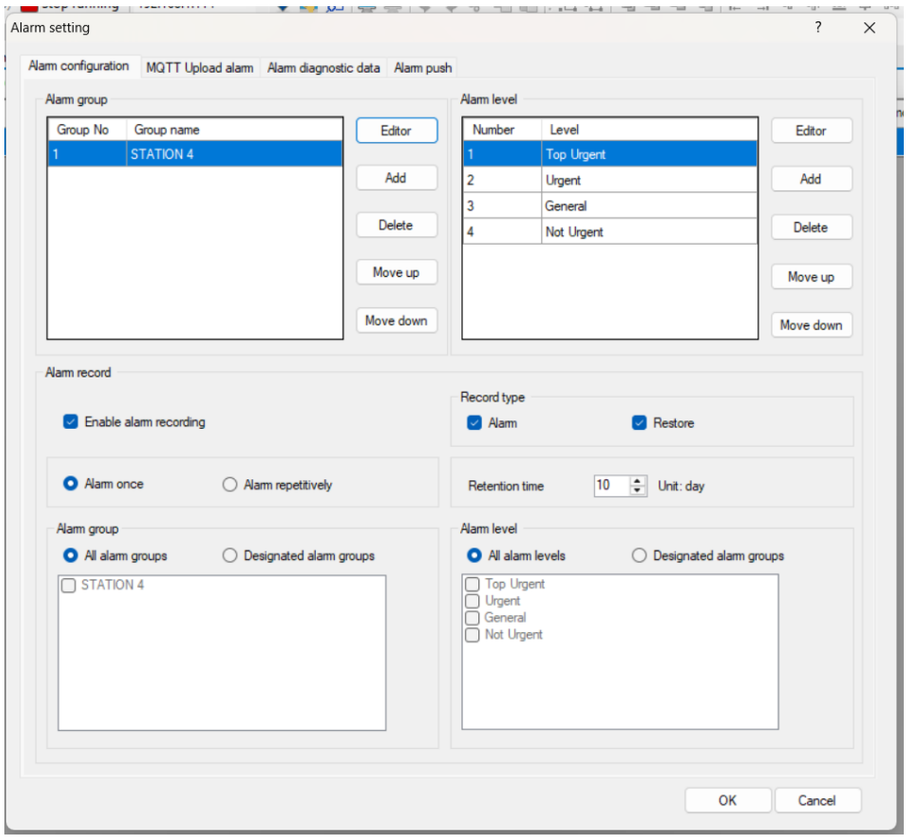
Symbol	Address	Comment
PAUSE_STATE	M0.7	PAUSE_STATE
PROCESS	M0.1	PROCESS START
STEP_1_DONE	M1.0	STEP_1_DONE
STEP_2_DONE	M1.1	STEP_2_DONE
STEP_3_DONE	M1.2	STEP_3_DONE
STEP_4_DONE	M1.3	STEP_4_DONE
STEP_5_DONE	M1.4	STEP_5_DONE
STEP_6_DONE	M1.5	STEP_6_DONE
STEP_7_DONE	M1.6	STEP_7_DONE
STEP_8_DONE	M1.7	STEP_8_DONE

15 RUNNING STATE



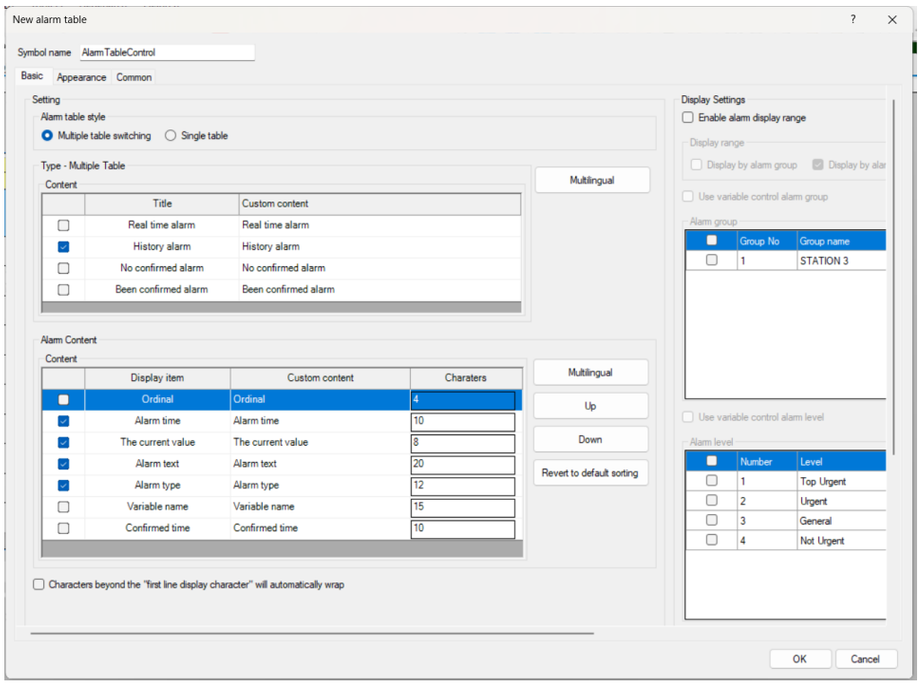
Symbol	Address	Comment
ON_STATE	M0.0	ON STATE
PAUSE_STATE	M0.7	PAUSE_STATE
PROCESS	M0.1	PROCESS START
RUNNING_STATE	M4.0	RUNNING STATE

13. Alarm Configuration



	Name	Alarm variable	Alarm condition	Alarm group	Alarm level	Screen push
1	EmergencyStopTriggered	Siemens_S7_200CN_SMART	Station 3 (System Halt):	STATION 3		☑

14. Alarm Management and Alarm History



Project Screenshots and Evidence



